

Lecture 6 Options

1. Introduction to Options

An **option** is a financial contract between a buyer and a seller that gives the buyer the right, but not the obligation, to buy or sell an underlying asset at a specified price on or before a specified date.

In this chapter, we focus on **European options**, which can only be exercised at maturity.

2. Call and Put Options

Call Option(看涨期权)

A **call option** gives the buyer the right to purchase the underlying asset at a predetermined price, called the **strike price** (K), on the expiration date.

Put Option (看跌期权)

A **put option** gives the buyer the right to sell the underlying asset at the strike price (K) on the expiration date.

3. Payoff Functions

Call Option Payoff

At expiration time t , the payoff of a call option is:

$$\text{Payoff} = \max(S(t) - K, 0) = (S(t) - K)^+$$

Put Option Payoff

At expiration time t , the payoff of a put option is:

$$\text{Payoff} = \max(K - S(t), 0) = (K - S(t))^+$$

4. Basic Option Terminology

- **Strike Price (执行价)** (K): the agreed-upon price for the underlying asset.
- **Spot Price(股票价格)** (S): current price of the underlying stock.
- **Expiration Date (到期日)** : date when the option contract expires.
- **Time to Expiration (到期期限)** (t): time remaining until expiration.

$$t = T - t_{\text{now}}$$

- **Premium(期权费)**: the price paid by the buyer to acquire the option contract.
- **In-the-Money (ITM)**:
 - 期权当前有内在价值，行权有利可图。
 - Call: $S > K$

- Put: $S < K$
- **Out-of-the-Money (OTM):**
 - 期权当前没有内在价值，不适合行权。
 - Call: $S < K$
 - Put: $S > K$
- Call = 未来我有权低价买入，所以我希望未来价格上涨。
- Put = 未来我有权高价卖出，所以我希望未来价格下跌。
- 无论 Call 还是 Put，最大亏损都不会超过买入合约时支付的 premium;
- 期权是一种典型的 非对称收益工具，适合用来做风险对冲或投机操作。

5. Profit and Loss (P&L) Diagrams

Call Option Buyer



Abbildung 1: Profit and Loss Diagram of a Call Option Buyer

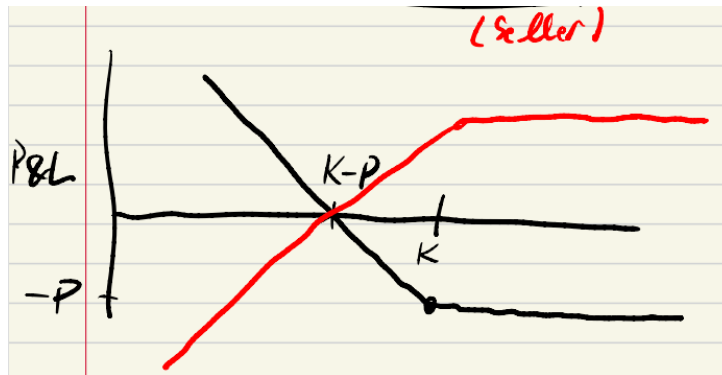
$$\text{Profit} = \begin{cases} S - K - P, & \text{if } S \geq K \text{ 行权获利} \\ -P, & \text{if } S < K \text{ 期权不行权, 损失为已支付的 premium} \end{cases}$$

Break-even point: $S = K + P$.

Call Option Seller



$$\text{Profit} = \begin{cases} P, & \text{if } S \leq K \\ P - (S - K), & \text{if } S > K \end{cases}$$



Put Option Buyer

$$\text{Profit} = \begin{cases} K - S - P, & \text{if } S \leq K \\ -P, & \text{if } S > K \end{cases}$$

Put Option Seller

$$\text{Profit} = \begin{cases} P, & \text{if } S \geq K \\ P - (K - S), & \text{if } S < K \end{cases}$$

6. Goal: Pricing Options

We want to compute the fair price of a European call/put option today.

$$C(0) = \mathbb{E} [(S(t) - K)^+]$$

当前价格等于未来所有可能收益的加权平均值，这里的加权就是“发生该情况的概率”。

$$P(0) = \mathbb{E} [(K - S(t))^+]$$

7. Modeling Stock Prices

We assume the stock price $S(t)$ follows **Geometric Brownian Motion (GBM)**:

$$S(t) = S(0)e^{\left(-\frac{\sigma^2}{2}t + \sigma\sqrt{t}Z\right)}$$

where:

- $Z \sim \mathcal{N}(0, 1)$ (standard normal distribution)
- σ is the volatility (standard deviation of log-returns)

8. The Black-Scholes Formula (期权定价公式)

Call Option Price (看涨期权价格)

$$C(0) = S(0)\Phi(d_1) - K\Phi(d_2)$$

- $\Phi(d_1), \Phi(d_2)$: 标准正态分布的累积分布函数 (CDF), 表示“涨上去的概率”。
- Call 买方的价值 = 股票有可能涨上去 $\times$ 涨上去时的收益 - 未来支付行权价的期望值。

Put Option Price (看跌期权价格)

$$P(0) = K\Phi(-d_2) - S(0)\Phi(-d_1)$$

Put 买方的价值 = 股票跌破行权价时的期望收益

where:

$$d_1 = \frac{\ln\left(\frac{S(0)}{K}\right) + \frac{\sigma^2}{2}t}{\sigma\sqrt{t}}, \quad d_2 = d_1 - \sigma\sqrt{t}$$

and $\Phi(\cdot)$ is the standard normal cumulative distribution function:

$$\Phi(y) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^y e^{-\frac{z^2}{2}} dz$$

- 数学上就是标准正态分布的累计概率函数 (累积分布函数)
- 例如 $\Phi(1.96) \approx 0.975$: 意味着正态分布下 97.5% 的概率落在这个点左侧

例子:定价一个看涨期权

- 当前股价: $S(0) = 100$
- 行权价: $K = 105$
- 波动率: $\sigma = 0.2$
- 到期时间: $t = 1$ 年
- 无风险利率暂时忽略 (即不贴现)

1. 计算 d_1, d_2 (衡量“股票价格超过行权价的可能性”的指标) :

$$d_1 = \frac{\ln(100/105) + \frac{0.2^2}{2} \cdot 1}{0.2 \cdot \sqrt{1}} = \frac{-0.0488 + 0.02}{0.2} = \frac{-0.0288}{0.2} \approx -0.144$$

$$d_2 = d_1 - \sigma\sqrt{t} = -0.144 - 0.2 = -0.344$$

2. 查表或计算: 通过“正态分布表”(Z-Table)查值。Z表给出了: 从负无穷到某个 z 值的累积概率 $\Phi(z)$

$$\Phi(d_1) \approx 0.443, \quad \Phi(d_2) \approx 0.366$$

```
[2]: from scipy.stats import norm
```

```
•[4]: # d1 和 d2 的数值
      d1 = -0.1439
      d2 = -0.344
```

```
•[6]: # 计算标准正态分布的CDF值
      phi_d1 = norm.cdf(d1)
      phi_d2 = norm.cdf(d2)
```

```
[8]: print(f"Phi(d1) = {phi_d1:.3f}")
      print(f"Phi(d2) = {phi_d2:.3f}")
```

```
Phi(d1) = 0.443
Phi(d2) = 0.365
```

3. 套用公式:

$$C(0) = S(0) \cdot \Phi(d_1) - K \cdot \Phi(d_2)$$

$$C(0) = 100 \cdot 0.443 - 105 \cdot 0.366 \approx 44.3 - 38.4 = 5.9$$

✓ 所以这个期权现在值 \$5.90。

9. Sketch of Black-Scholes Derivation

使用 Indicator 函数拆分期望, We use:

$$\mathbb{E}[(S(t) - K)^+] = \mathbb{E}[(S(t) - K)1_{S(t) \geq K}]$$

- 因为 $(S(t) - K)^+$ 其实就是当 $S(t) \geq K$ 时才有值, 否则为 0;
- 所以我们可以用 Indicator 函数 $1_{\{S(t) \geq K\}}$ 把它表达出来。

This can be rewritten as (拆成两个期望) :

$$\mathbb{E}[(S(t) - K)^+] = \mathbb{E}[S(t) \cdot 1_{S(t) \geq K}] - K\mathbb{E}[1_{S(t) \geq K}]$$

- $S(t) \cdot 1$: 只在“股票涨过 K ”时考虑股票的价格;
- $K \cdot 1$: 只在“涨过 K ”时才减去行权价。

The two expectations are computed separately, yielding:

$$\mathbb{E} [S(t) \cdot 1_{S(t) \geq K}] = S(0)\Phi(d_1)$$

$$\mathbb{E} [1_{S(t) \geq K}] = \Phi(d_2)$$

1. 第一个期望:

$$\mathbb{E} [S(t) \cdot \mathbf{1}_{\{S(t) \geq K\}}] = S(0)\Phi(d_1)$$

- 这是未来股票价格超过 K 时的加权平均值
- 实际上利用了股票价格服从几何布朗运动，再转为正态分布后积分得出（涉及风险中性概率测度）

2. 第二个期望:

$$\mathbb{E} [\mathbf{1}_{\{S(t) \geq K\}}] = \Phi(d_2)$$

- 这是股票价格涨过 K 的“概率”
- 就是 $Z > \text{某个数}$ 的概率，直接查标准正态表得到

Thus,

$$C(0) = S(0)\Phi(d_1) - K\Phi(d_2)$$

Problem Setup

We assume that the stock price $S(t)$ at expiration time t follows a Geometric Brownian Motion (GBM):

$$S(t) = S(0)e^{-\frac{\sigma^2}{2}t + \sigma\sqrt{t}Z},$$

where $Z \sim \mathcal{N}(0, 1)$ is a standard normal random variable, and σ is the volatility.

We want to compute the value of the European call option:

$$C(0) = \mathbb{E} [(S(t) - K)^+],$$

where K is the strike price and $(x)^+ = \max(x, 0)$.

Splitting the Expectation

We rewrite the expectation as:

$$\mathbb{E} [(S(t) - K)^+] = \mathbb{E} [(S(t) - K) \cdot 1_{\{S(t) \geq K\}}].$$

Using linearity of expectation:

$$\mathbb{E} [(S(t) - K)^+] = \mathbb{E} [S(t) \cdot 1_{\{S(t) \geq K\}}] - K\mathbb{E} [1_{\{S(t) \geq K\}}].$$

Thus we need to compute two parts:

- $\mathbb{E} [1_{\{S(t) \geq K\}}]$
- $\mathbb{E} [S(t) \cdot 1_{\{S(t) \geq K\}}]$

Computing $\mathbb{E} [1_{\{S(t) \geq K\}}]$

We start by solving for the condition $S(t) \geq K$:

$$S(0)e^{-\frac{\sigma^2}{2}t + \sigma\sqrt{t}Z} \geq K.$$

Taking logarithms:

$$-\frac{\sigma^2}{2}t + \sigma\sqrt{t}Z \geq \ln\left(\frac{K}{S(0)}\right).$$

Rearranging:

$$Z \geq \frac{\ln\left(\frac{K}{S(0)}\right) + \frac{\sigma^2}{2}t}{\sigma\sqrt{t}}.$$

Define:

$$d_2 = \frac{\ln\left(\frac{S(0)}{K}\right) - \frac{\sigma^2}{2}t}{\sigma\sqrt{t}}.$$

Thus:

$$\mathbb{P}(S(t) \geq K) = \mathbb{P}(Z \geq d_2) = 1 - \Phi(d_2).$$

So:

$$\mathbb{E} [1_{\{S(t) \geq K\}}] = \Phi(d_2).$$

Computing $\mathbb{E} [S(t) \cdot 1_{\{S(t) \geq K\}}]$

We rewrite:

$$\mathbb{E} [S(t) \cdot 1_{\{S(t) \geq K\}}] = S(0)\mathbb{E} \left[e^{-\frac{\sigma^2}{2}t + \sigma\sqrt{t}Z} \cdot 1_{\{Z \geq d_2\}} \right].$$

Factor out $S(0)$:

$$= S(0)e^{-\frac{\sigma^2}{2}t} \mathbb{E} \left[e^{\sigma\sqrt{t}Z} \cdot 1_{\{Z \geq d_2\}} \right].$$

Note that $e^{\sigma\sqrt{t}Z}$ is log-normal, and rewrite:

$$e^{\sigma\sqrt{t}Z} = e^{\frac{\sigma^2 t}{2}} \cdot e^{\sigma\sqrt{t}(Z - \sigma\sqrt{t}/2)}.$$

Thus,

$$\mathbb{E} [S(t) \cdot 1_{\{S(t) \geq K\}}] = S(0)\mathbb{E} \left[e^{\sigma\sqrt{t}Z} \cdot 1_{\{Z \geq d_2\}} \right] = S(0)\Phi(d_1),$$

where

$$d_1 = \frac{\ln\left(\frac{S(0)}{K}\right) + \frac{\sigma^2}{2}t}{\sigma\sqrt{t}} = d_2 + \sigma\sqrt{t}.$$

Black-Scholes 模型中定义:

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}} \quad d_2 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r - \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}}$$

我们将 d_2 与 d_1 相减:

$$\begin{aligned} d_1 - d_2 &= \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}} - \frac{\ln\left(\frac{S_0}{K}\right) + \left(r - \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}} \\ &= \frac{\left[\left(r + \frac{\sigma^2}{2}\right)t - \left(r - \frac{\sigma^2}{2}\right)t\right]}{\sigma\sqrt{t}} \\ &= \frac{\left(\frac{\sigma^2}{2} + \frac{\sigma^2}{2}\right)t}{\sigma\sqrt{t}} \\ &= \frac{\sigma^2 t}{\sigma\sqrt{t}} \\ &= \sigma\sqrt{t} \end{aligned}$$

因此我们得到:

$$d_2 = d_1 - \sigma\sqrt{t}$$

这个结果说明, d_2 比 d_1 少一个“风险调整后的时间不确定性项”, 也体现了资产在未来的不确定性随着波动率 and 时间的增长而增加。

- d_1 : 考虑了“现在”从当前价格上涨到行权价的距离 + 一个偏移项 ($\frac{1}{2}\sigma^2$)
- d_2 : 去掉了那部分偏移, 仅考虑风险中性下未来是否行权的概率

Conclusion

Now putting both pieces together:

$$C(0) = S(0)\Phi(d_1) - K\Phi(d_2).$$

Put Option Formula (Put-Call Parity)

The formula for the European put option can be derived using put-call parity:

$$C(0) - P(0) = S(0) - K.$$

Thus:

$$P(0) = C(0) + K - S(0) = -S(0)\Phi(-d_1) + K\Phi(-d_2).$$

Summary of Black-Scholes Formula

$$\begin{aligned}C(0) &= S(0)\Phi(d_1) - K\Phi(d_2), \\P(0) &= K\Phi(-d_2) - S(0)\Phi(-d_1),\end{aligned}$$

with

$$d_1 = \frac{\ln\left(\frac{S(0)}{K}\right) + \frac{\sigma^2}{2}t}{\sigma\sqrt{t}}, \quad d_2 = d_1 - \sigma\sqrt{t}.$$

Standard Normal CDF

$$\Phi(y) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^y e^{-z^2/2} dz.$$

10. Call-Put Parity

The relationship between call and put prices is:

$$C(0) - P(0) = S(0) - K$$

This allows us to compute one option price from the other.

这是欧式期权最核心的无套利关系:

$$C - P = S_0 - Ke^{-rt}$$

其中:

- C : 看涨期权 (Call) 价格
- P : 看跌期权 (Put) 价格
- S_0 : 当前标的资产价格
- K : 行权价格
- r : 无风险利率
- t : 到期时间

✓ Put 和 Call 的价格不能随便偏离, 否则会存在套利空间。

11. Remarks

- Option prices incorporate expectations of future volatility and market behavior.
- Selling options carries large risk if not managed properly.
- The Black-Scholes model assumes:
 - No arbitrage
 - Constant interest rate (we assumed $r = 0$ here)
 - Constant volatility
 - Log-normal stock price distribution

Tabelle 1: Black-Scholes 与 Monte Carlo Simulation 方法比较

对比维度	Black-Scholes	Monte Carlo Simulation
方法类别	解析解 (Closed-form Solution)	数值解 (Numerical Method)
核心思想	利用偏微分方程推导出的公式计算欧式期权价格	模拟大量可能的资产路径, 计算平均 payoffs 并贴现回现值
精度	理论精确, 只受浮点误差影响	有统计误差, 精度依赖于模拟路径数量 (样本数量)
适用范围	只适用于欧式期权, 假设为 GBM (几何布朗运动), 不能处理路径异常路径或 path-dependent 产品	几乎适用于所有路径依赖、复杂 payoff, 甚至美式期权等
计算效率	快速 (一个公式即可)	慢, 需大量随机模拟路径, 尤其在多维问题中
易于可视化 / 直观性	难以可视化	容易展示路径, 适合教学、直觉理解
可扩展性 (跳跃、局部波动率等)	差, 需要重新推导 PDE 或封闭解	强, 可直接在模拟模型中修改过程, 如 Heston、Jump-Diffusion、Stochastic Volatility 等

Black-Scholes 核心原理回顾:

Black-Scholes 模型基于以下假设:

- 标的资产遵循几何布朗运动 (Geometric Brownian Motion) :

$$dS = \mu S dt + \sigma S dW_t$$

- 利用风险中性定价原理, 结合偏微分方程 (PDE) 求解, 得到欧式期权定价公式:

$$C(0) = S_0 \Phi(d_1) - K e^{-rt} \Phi(d_2)$$

Monte Carlo Simulation 核心原理:

Monte Carlo 方法不需要解析公式, 它只需要模拟 “现实” 中资产价格可能走的路径:

- 在风险中性世界中, 使用:

$$S_T = S_0 \cdot \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) t + \sigma \sqrt{t} Z \right), \quad Z \sim \mathcal{N}(0, 1)$$

- 生成大量路径 $S_T^{(i)}$
- 计算对应 payoffs, 例如 Call 为 $\max(S_T - K, 0)$

4. 求平均后折现:

$$C(0) \approx e^{-rt} \cdot \frac{1}{N} \sum_{i=1}^N \max(S_T^{(i)} - K, 0)$$